

PROIECTAREA CU MICROPROCESOARE

C3 2025-2026 Sem. 2

Proiect 6

Sapt 6

1 aprilie 2026

19:00-20:00

serban@upit.ro

Regulament disciplina

Nota finala este formata din activitatile:

- Laborator 15%
- Lucrare control 15%
- Prezentare si sustinere proiect 20%
- Examen 50%
- Bonus prezenta – 10%

Conditii pentru promovare:

- Nota de la laborator trebuie sa fie minim 5 (prezenta obligatorie la toate sedintele de laborator);
- Nota proiect minim 5 (conditie obligatorie pentru intrare in examen);
- Nota de la examen trebuie sa fie minim 5;
- Nota de minim 5 la Lucrarea de control.

In cazul reluarii disciplinei intr-un alt an universitar, activitatile nepromovate trebuie parcurse din nou.

Cerinte proiect

1). Realizarea suportului scris (format pdf) care sa contina capitolele:

Proiectarea Hardware si Proiectarea Software

- In cap. Proiectarea Hardware trebuie sa fie inclusa argumentarea pentru logica de conectare a circuitelor de memorie si I/O, respectiv a perifericelor;
- Este **obligatorie** schema de conectare finala a tuturor componentelor (eventual fara circuitele de alimentare) cu precizarea tuturor pinilor circuitelor folosite si a modului de conectare a acestora; va fi prezentat modul de programare a tablei prin folosirea tastelor (manualul de utilizare);
- In cap. Proiectarea Software trebuie sa fie inclusa logica de constructie a programului software cu toate componentele sale (prin descrieri explicative si organigrame), respectiv listingul programului;
- Este **obligatorie** lista de variabile folosite sub forma unui tabel de forma:

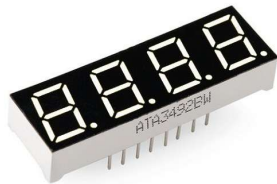
Nume variabila	Semnificatie	Valoarea de initializare	Mod de reprezentare	Adresa de stocare in memorie RAM

- Folosirea unui simulator pentru programul (sau portiuni din el) realizat este un bonus.

2). Predarea proiectului se va face pana in ziua de **23 mai 2026 ora 23:00** pe platforma moddle.

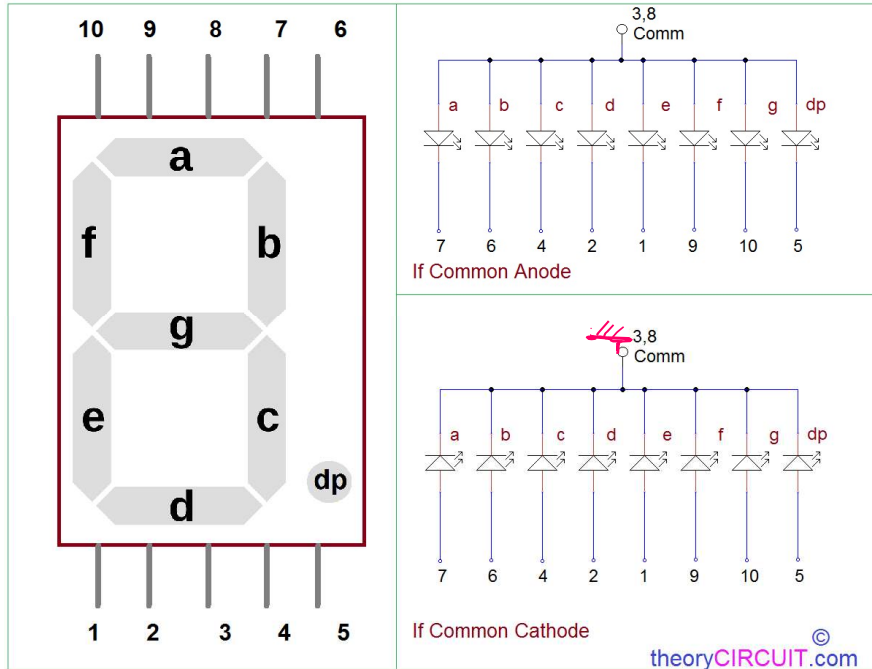
3). Sustinerea proiectului de catre realizator (activitate obligatorie) printr-o sesiune de intrebari legate de solutiile folosite (in ultima saptamana din semestru 25 – 29 mai 2026 la o data stablita de comun acord).

Celule de afisaj LED cu 7 segmente



Celule de afisaj LED cu 7 segmente

7 Segment Display Pinout

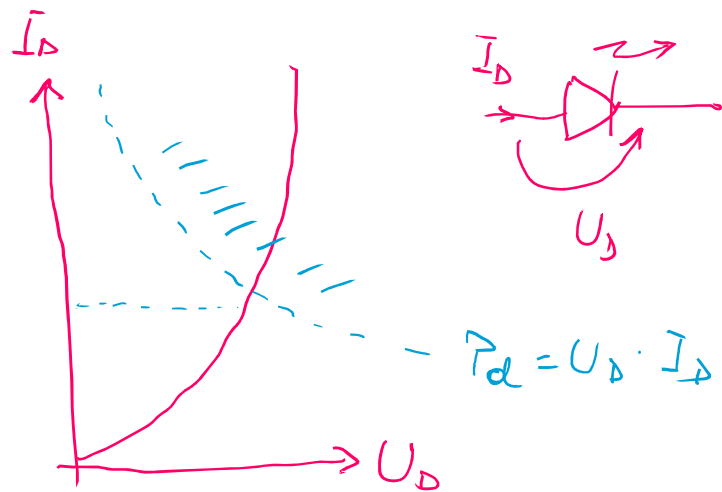
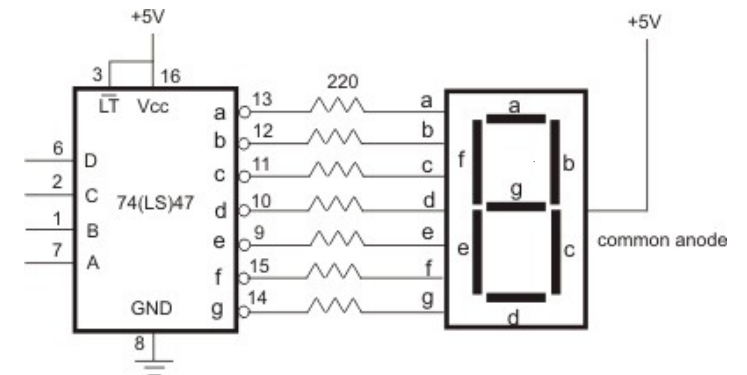
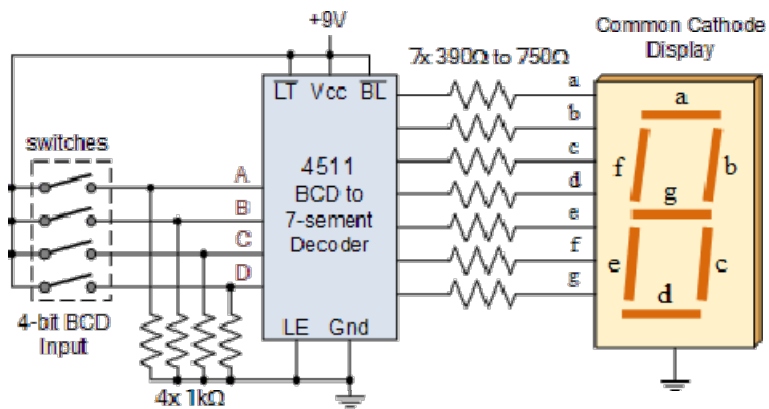


CATHODE COMMON

segment outputs							display
a	b	c	d	e	f	g	
1	1	1	1	1	1	0	0
0	1	1	0	0	0	0	1
1	1	0	1	1	0	1	2
1	1	1	1	0	0	1	3
0	1	1	0	0	1	1	4
1	0	1	1	0	1	1	5
0	0	1	1	1	1	1	6
1	1	1	0	0	0	0	7
1	1	1	1	1	1	1	8
1	1	1	0	0	1	1	9

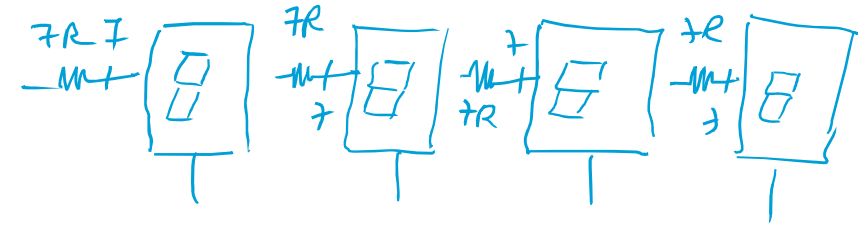
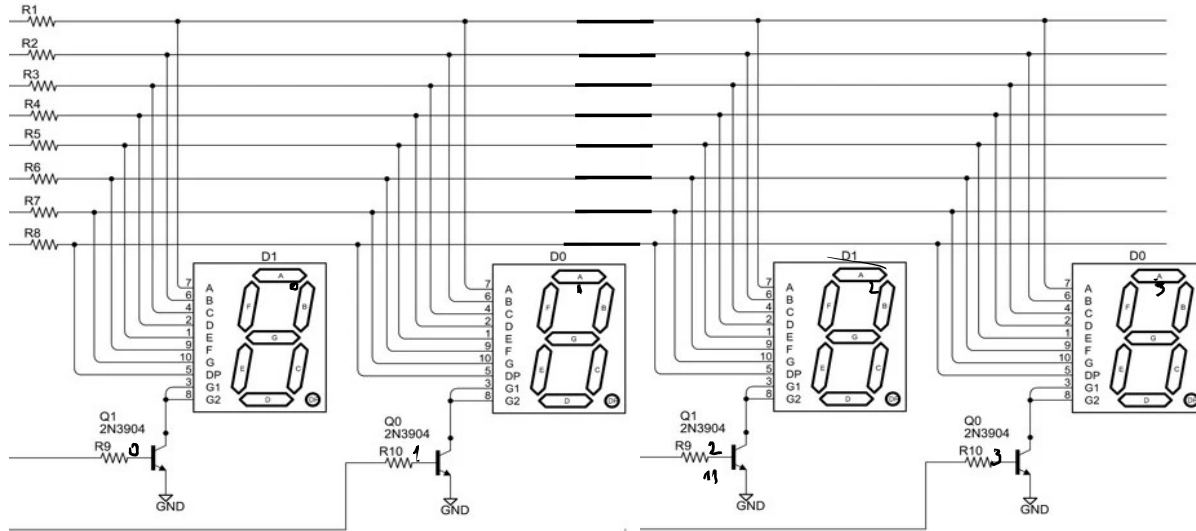
7-Segment Display

Celule de afisaj LED cu 7 segmente



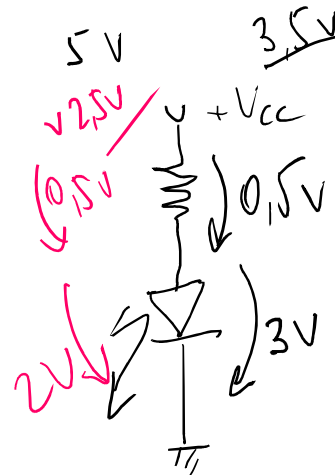
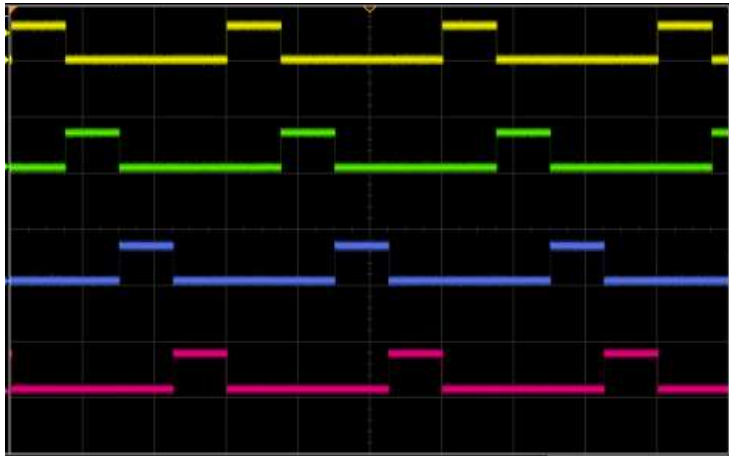
comanda clasică pentru
celula LED 7 segmente.

Proiectarea afisajului cu celule LED 7 segmente comandat prin multiplexare



$7R \times 4 = 28R$ Rezistente
 $7 \times 4 = 28$ fire comanda
 4 - catod
 (comanda clasic)

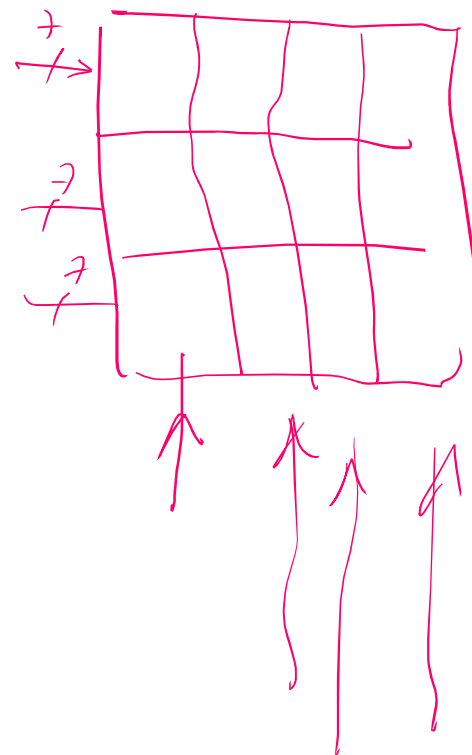
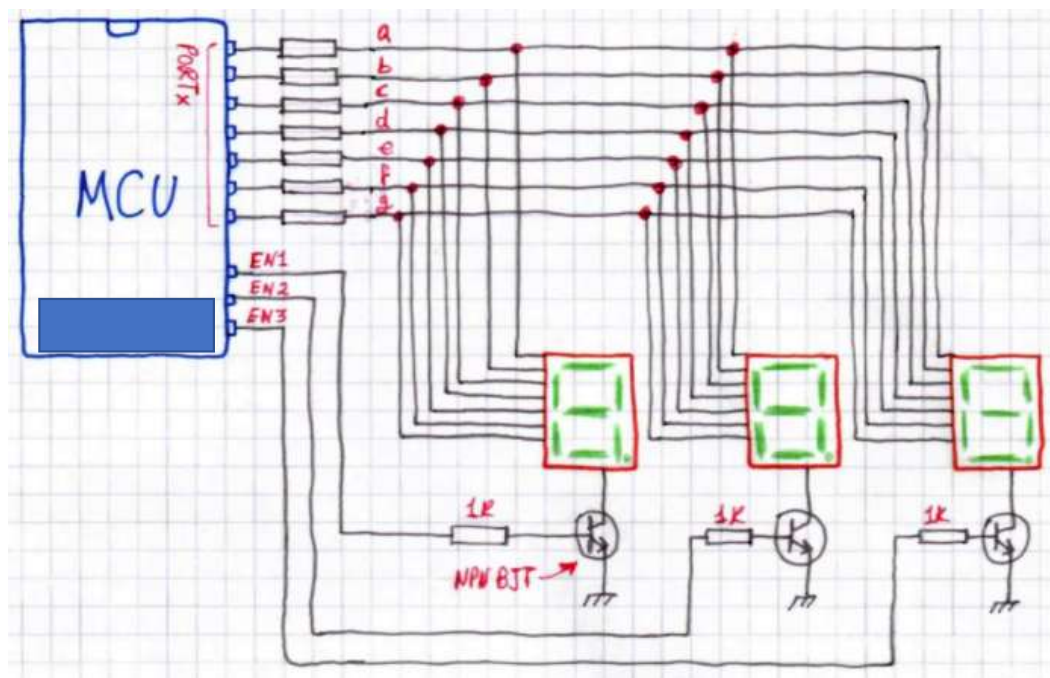
$t = ?$
 $f > 50 \text{ kHz}$ $T < 20 \text{ ms}$

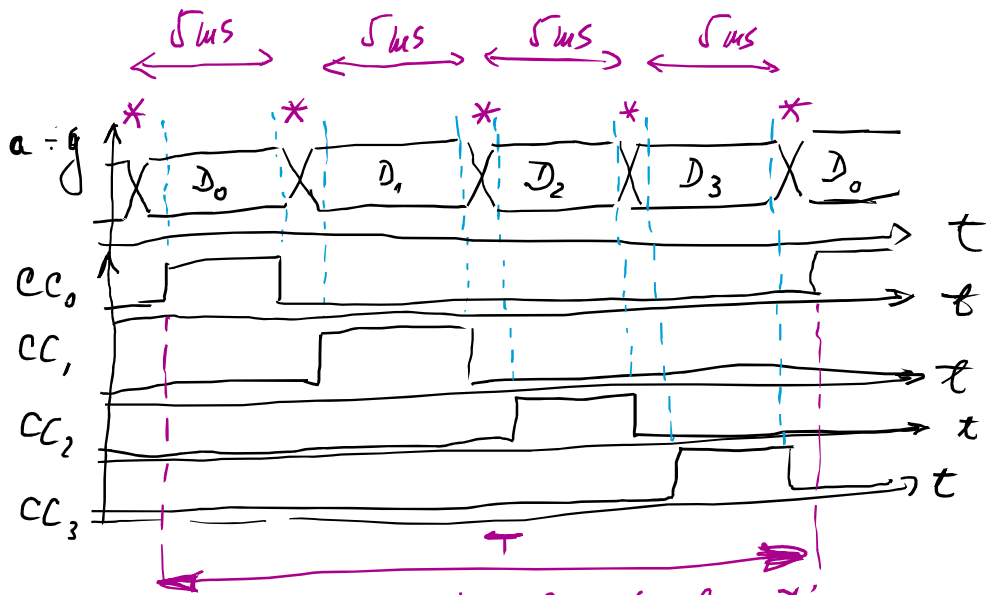


Comanda prin multiplexare
 Resurse

- $7R$ - rezistente
 - $7 + 4 = 11$ fire comanda

Proiectarea afisajului cu celule LED 7 segmente comandat prin multiplexare



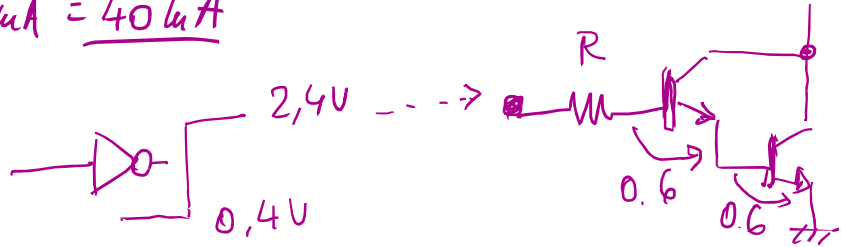


$T \leq 20 \mu s$ ($f \geq 50 \text{ kHz}$) $n = 4$ digits
 $T/n = \frac{20 \mu s}{4} = 5 \mu s \Rightarrow$ necesitate folosire timer

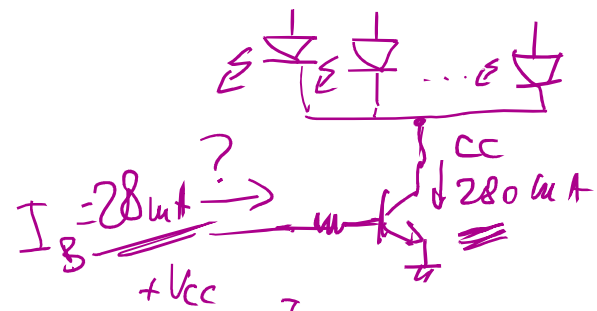
$\Rightarrow$ semnalizari la 5 μs prin cereri de intrerupere către CPU

$$I_{peak} \geq n \cdot I_D = 4 \cdot 10 \mu A = 40 \mu A$$

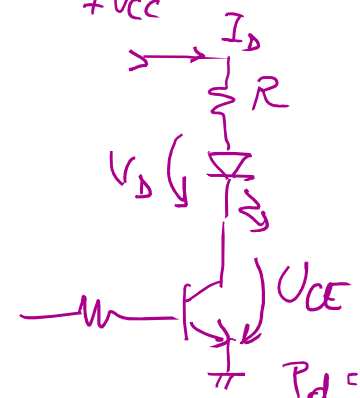
$\Rightarrow$ necesitate circuite amplificare curent



Circuitul de amplificare în curent Ct_2 (cotelod comun) -



medie putere BD 141, 143



$$V_{CC} = I_D \cdot R + V_D + U_{CE}$$

$$5 = 40 R + 2 + \frac{U_{CE_{SAT}}}{1V} \quad (2N2222)$$

$$R = \frac{5 - 2 - 1}{40} = \frac{2}{40} = 0.05 k = 50 \Omega$$

$$P_d = U_{CE} \cdot I_C$$

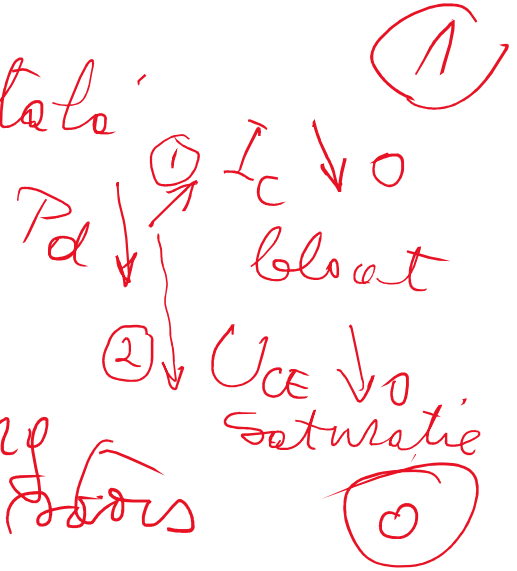
$$\beta_{SAT} = 10$$

Regim saturat $U_{CE} \rightarrow 0$
 Regim blocat $I_C \rightarrow 0$

Operare cu tranzistori in logica digitala

1) Putere disipata-

$$P_{d\text{tranz}} = U_{CE} \cdot I_C$$



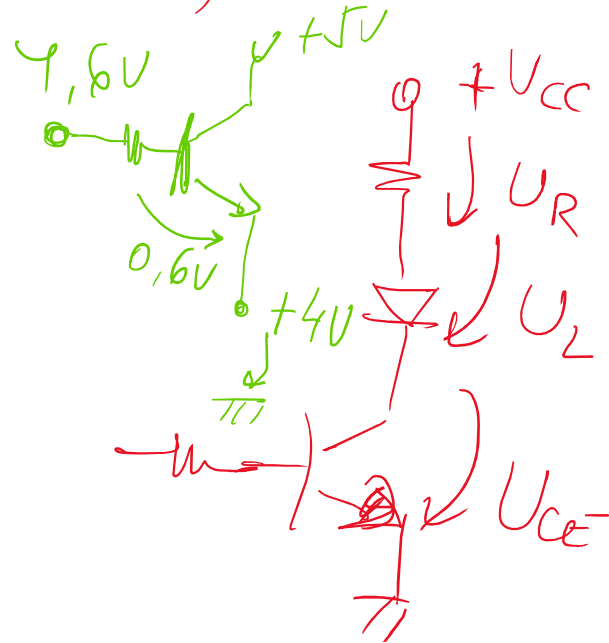
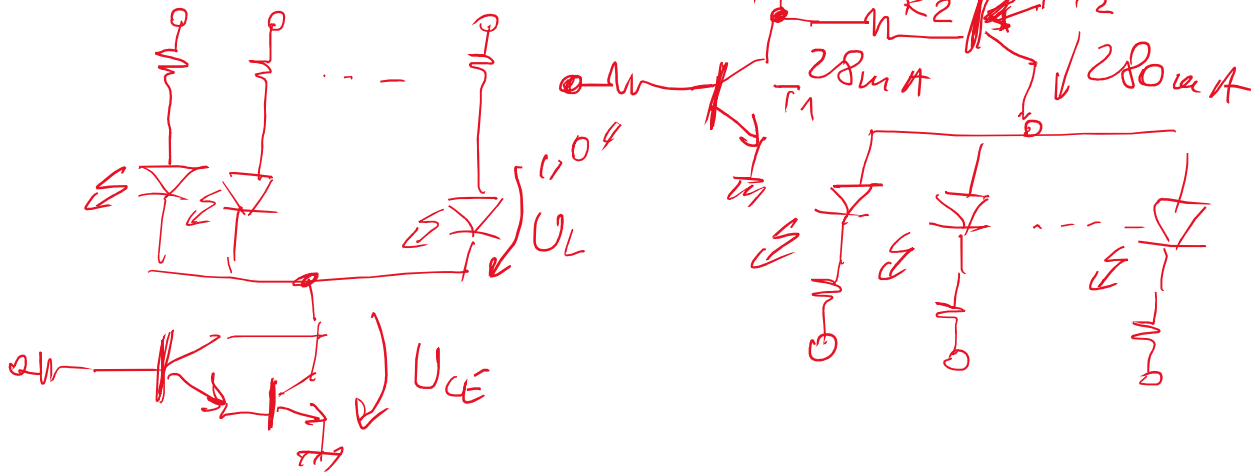
Switching transistors

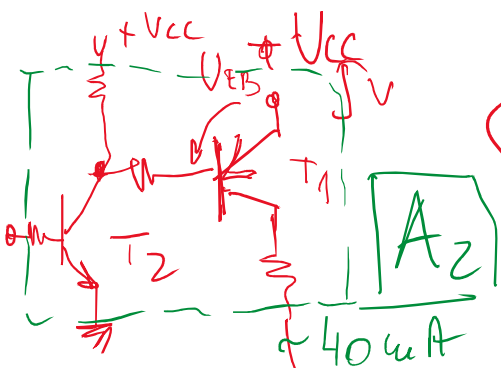
$$P = U \cdot I = I \cdot R \cdot I = I^2 \cdot R$$



Catod comun

Anod comun

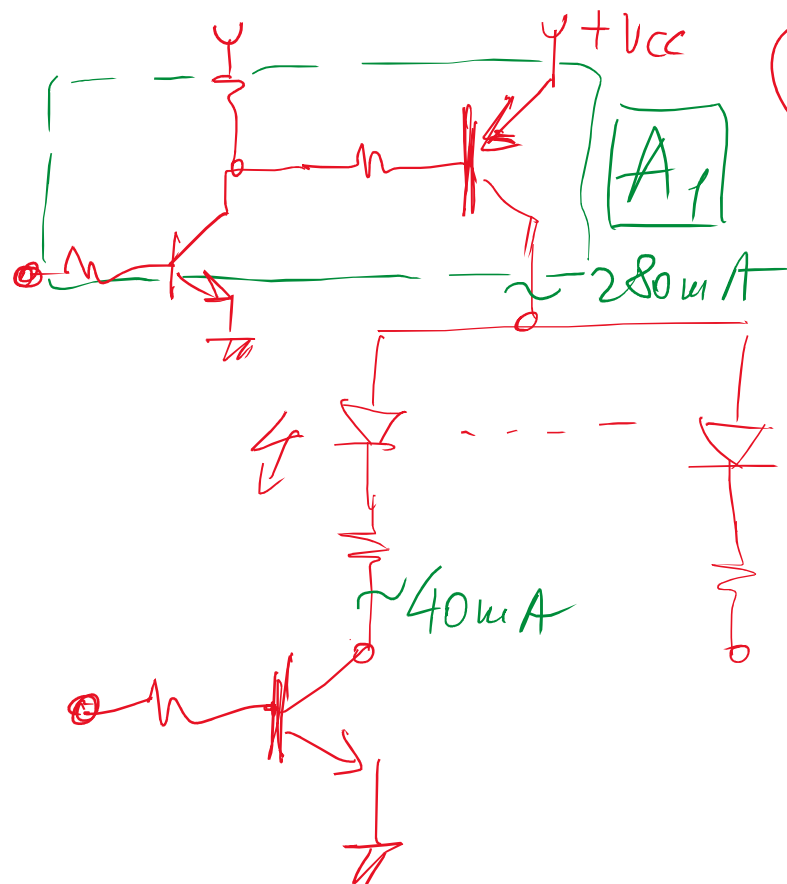
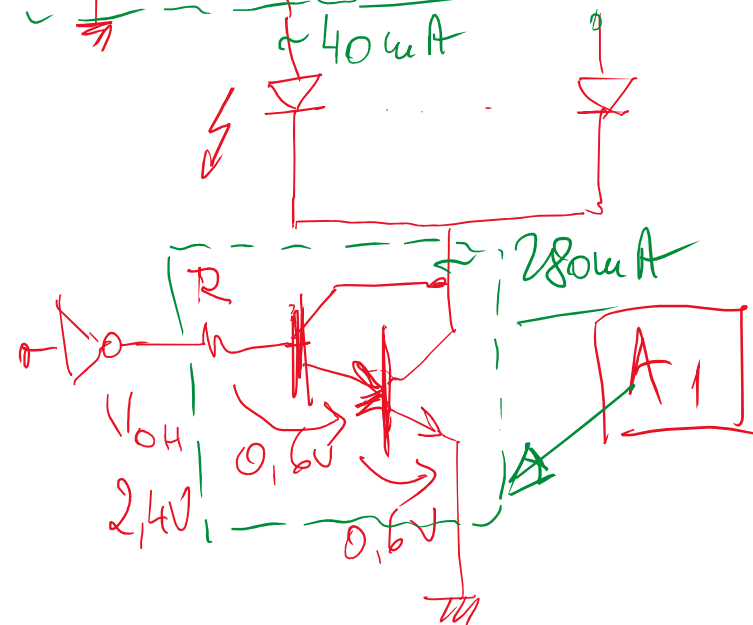




(CC)

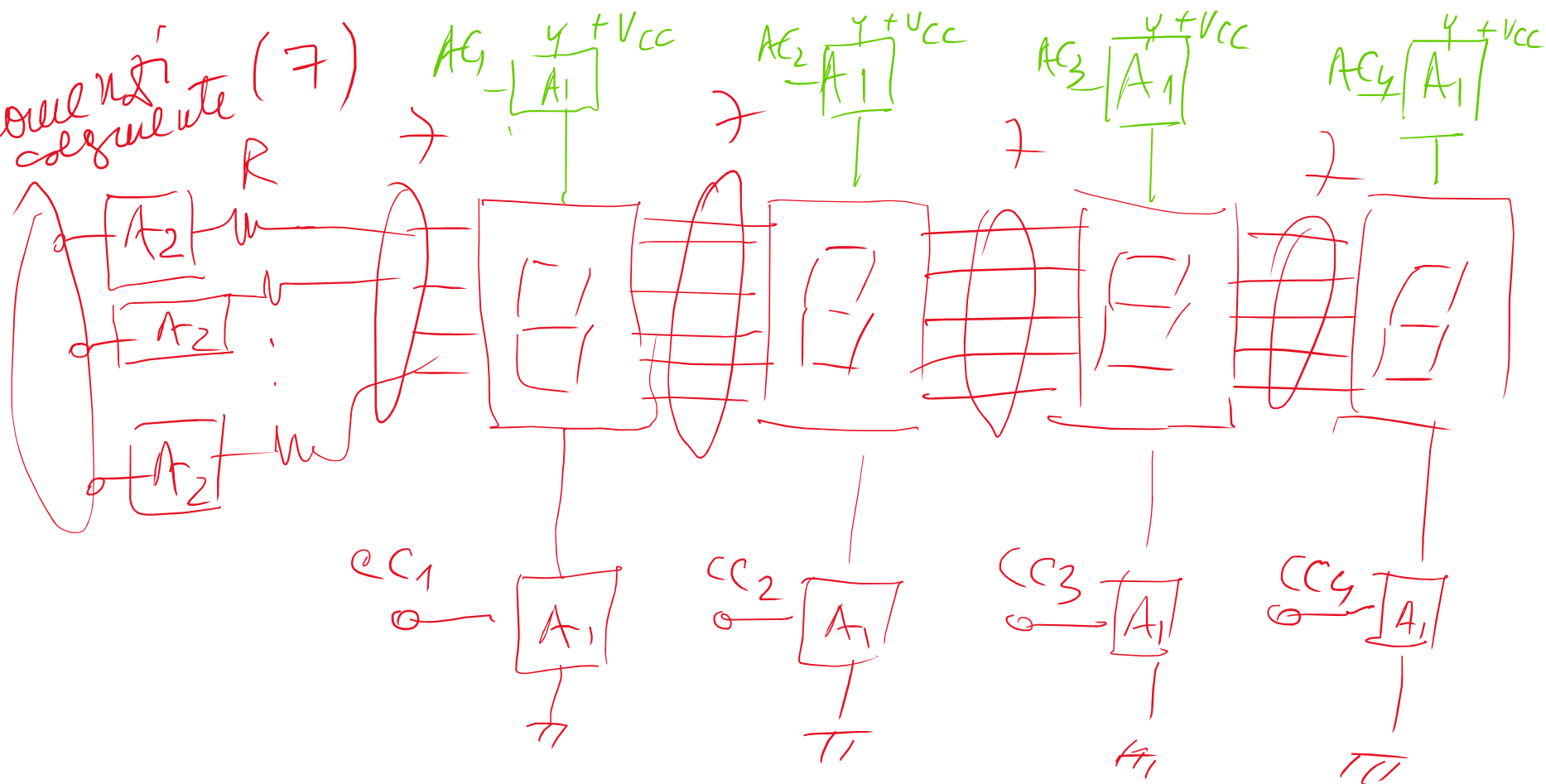
$$V_{OL} \leq 0,4V$$

$$V_{OH} \geq 2,4V$$



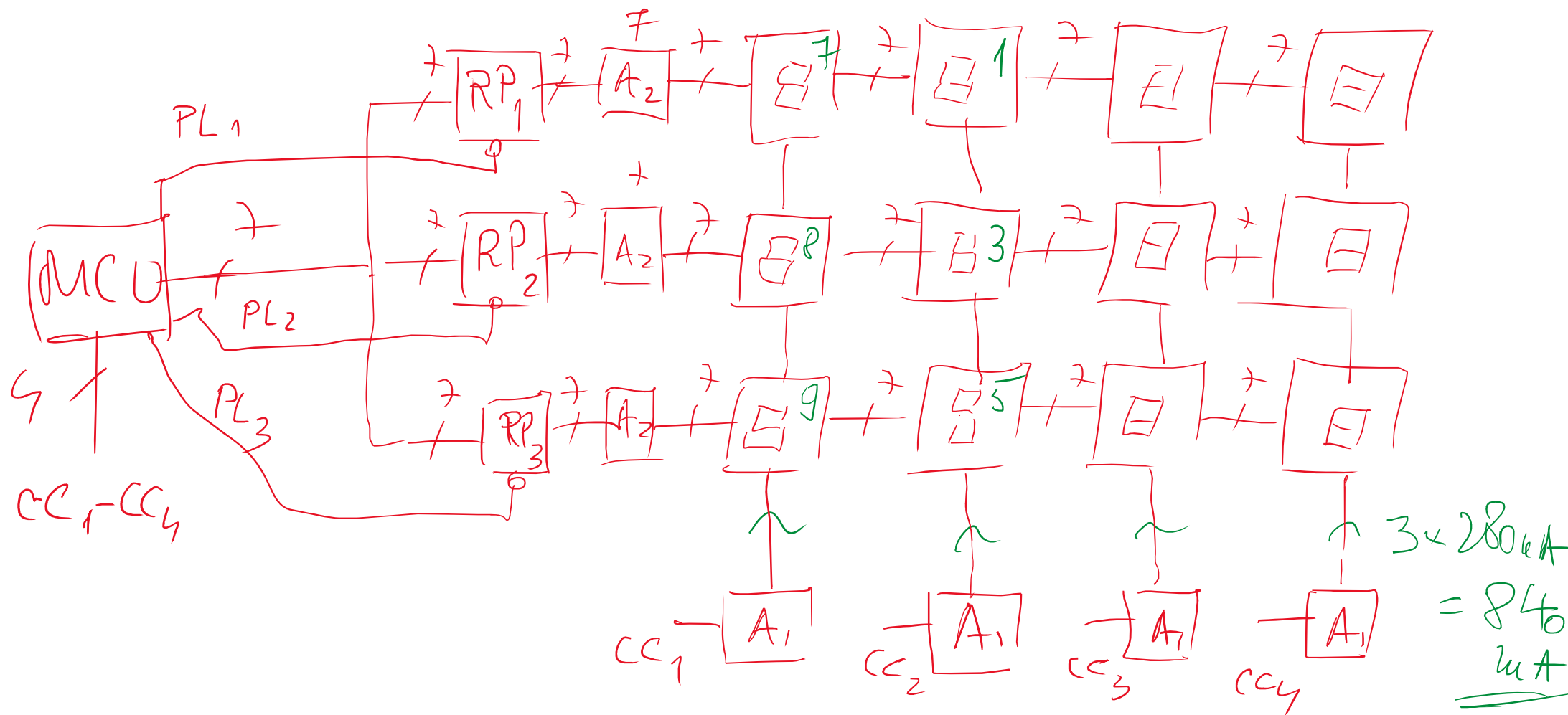
(AC)

could not
calculate (7)



AE

PL-parallel load

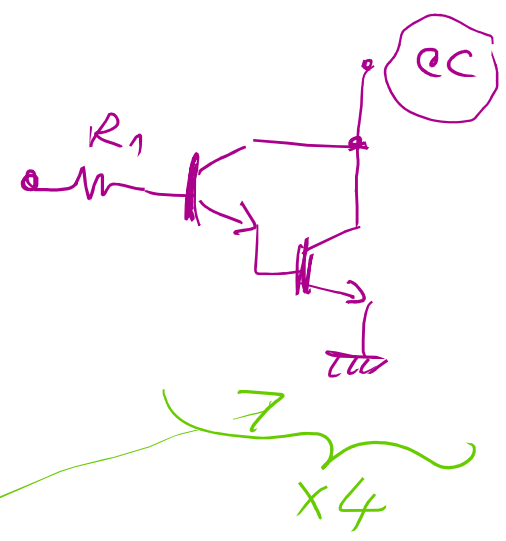
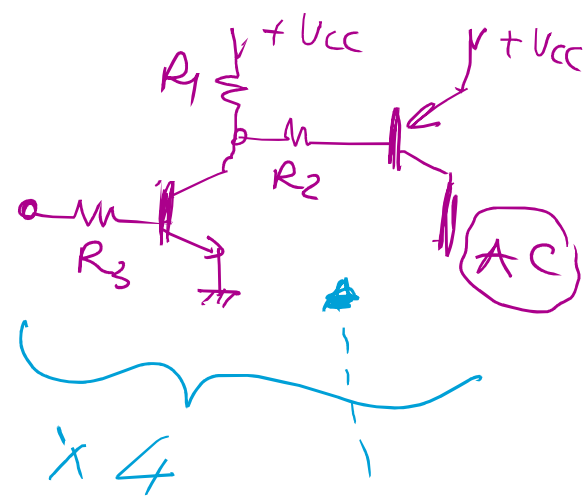
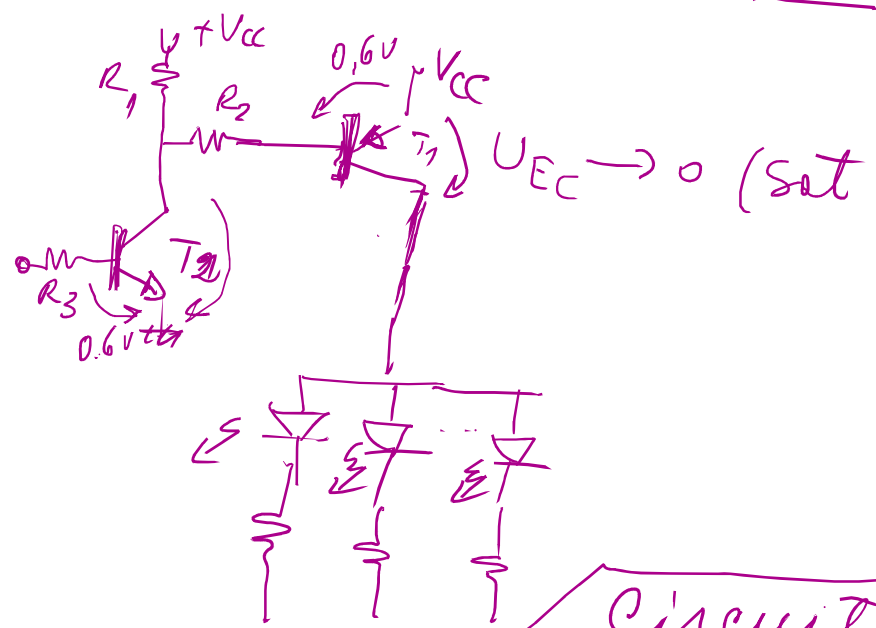


SUBTEME

- 1) Căutare celule LED 7seg (www.farnell.com)
- 3) Căutare tranzistoare (—||—) pentru A_1 și A_2
- 2) Scheme principale de multiplexare pentru matricea de celule 7seg din tablă
(calcul curent)
- 4) Proiectarea A_1 și A_2 (valori de rezistențe + alegere tranzistor)

CA2

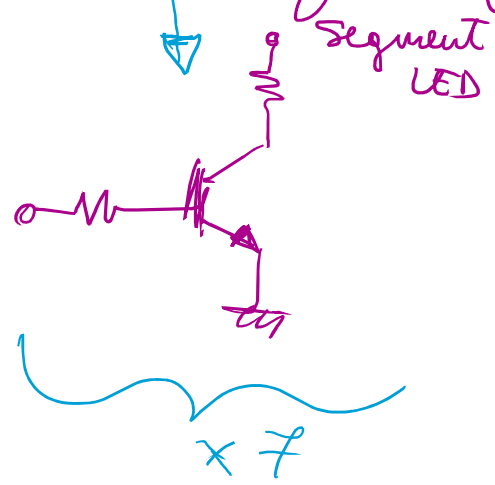
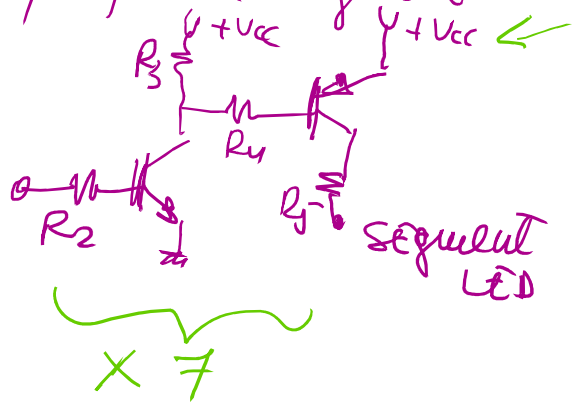
ANOD COMUN

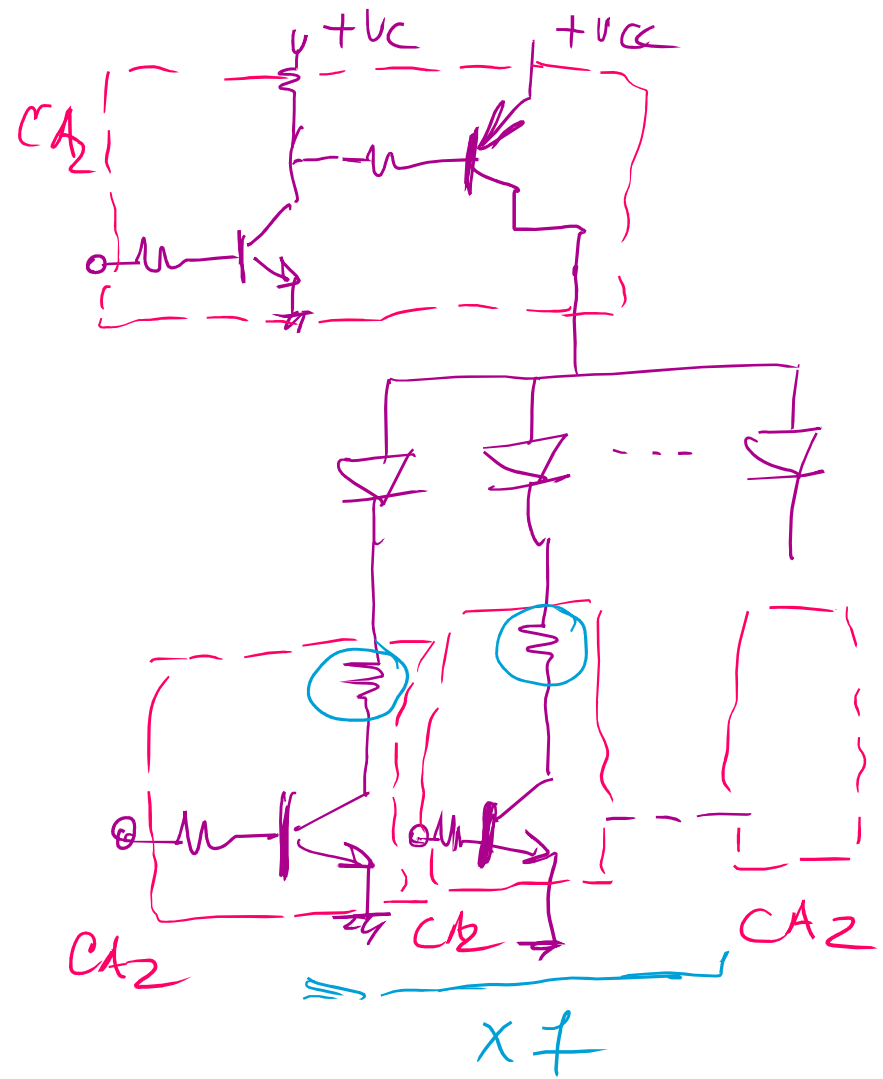
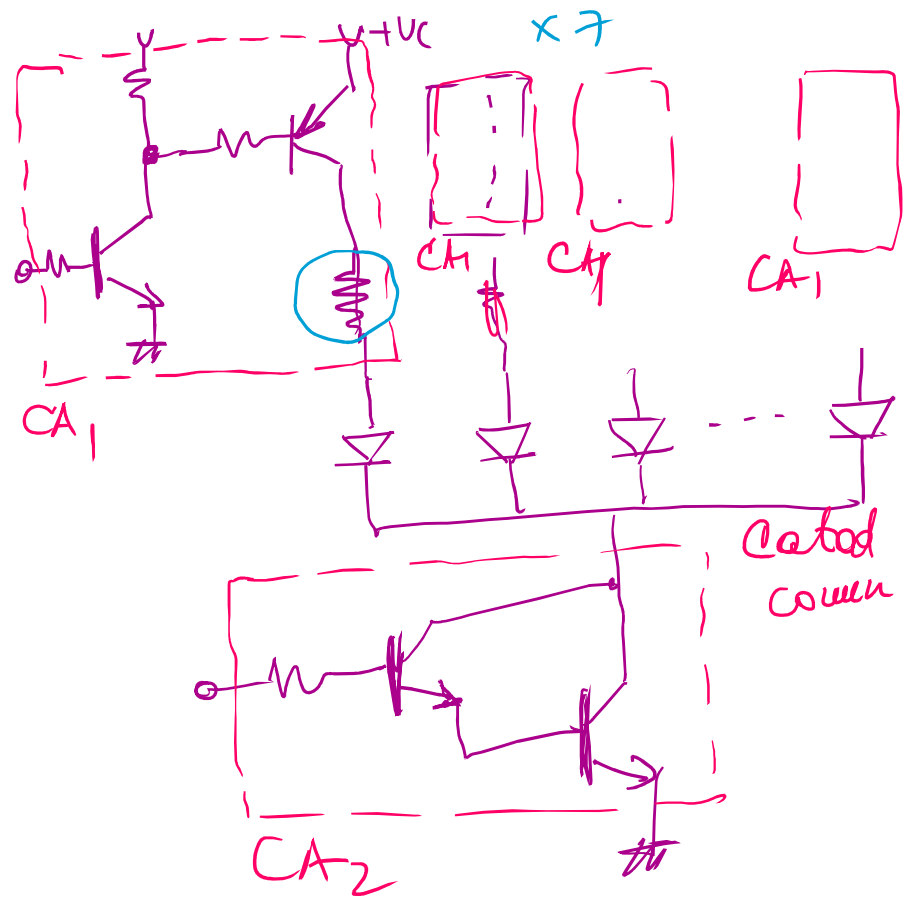


Circuit amplificare curent pe segmente (CA₁)

CATOD COMUN

pe fiecare segment



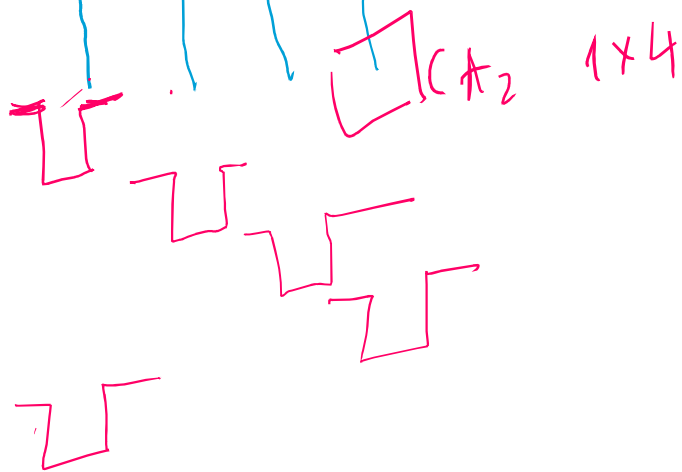
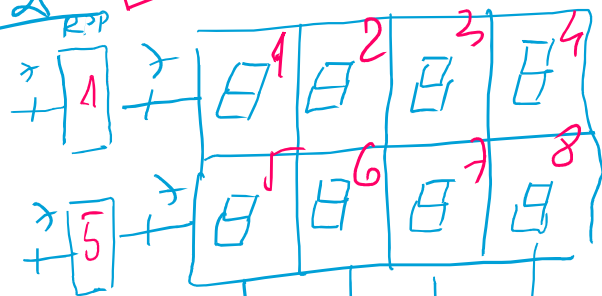


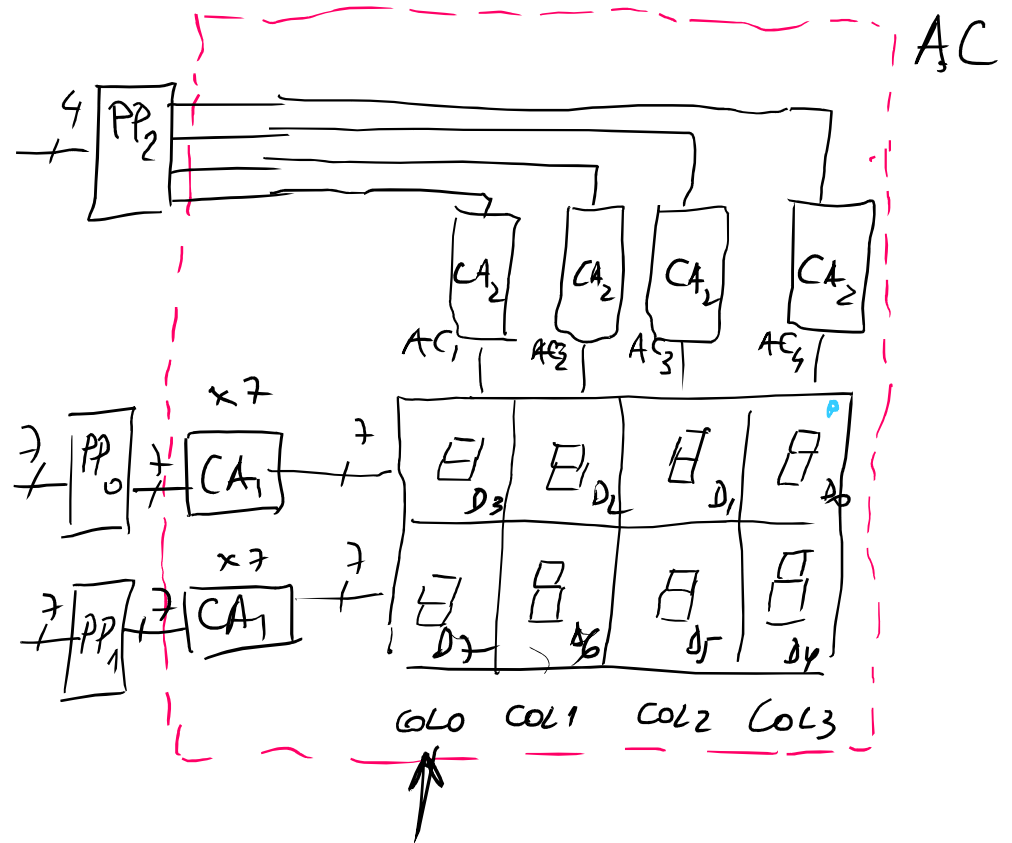
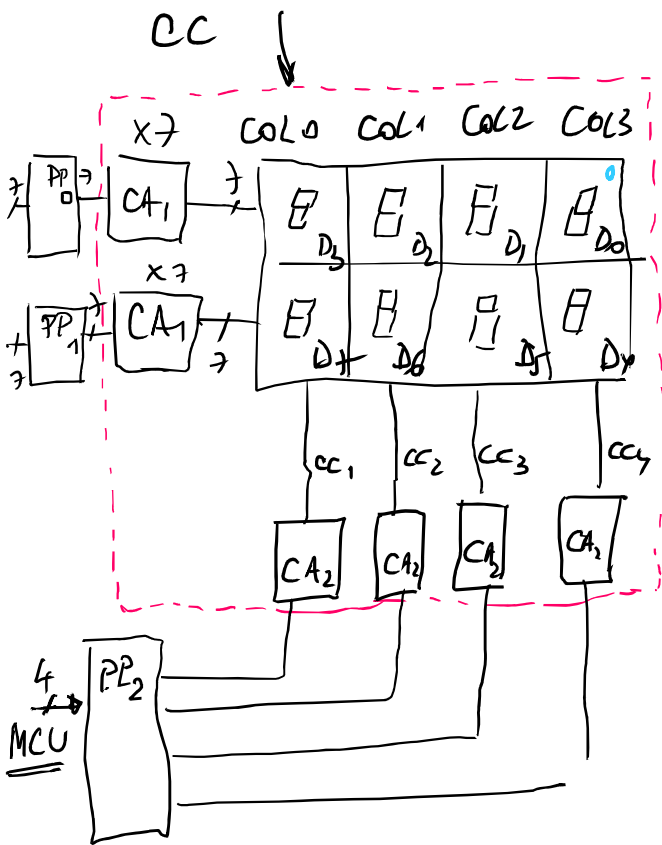
Dispunere celule afroaj

caz 1



caz 2 $\square$ CA1 $\rightarrow 2 \times 4$





END 4 mar 2026

D ₃	D ₂	D ₁	D ₀
D ₇	D ₆	D ₅	D ₄

(comando
electrice)

D ₃	D ₄	D ₅	D ₆	D ₇	D ₈	D ₉	D ₀
----------------	----------------	----------------	----------------	----------------	----------------	----------------	----------------

t_0 1 3 5 9 7 8 4 0
 t_1 3 5 9 7 8 4 0 1
 t_2 5 9 7 8 4 0 1 3

RAM

CONV.
 BIN
 7 SG

⇒
 programme
 a fisare

D ₀
D ₁
D ₂
D ₃
D ₄
D ₅
D ₆
D ₇
DBIN ₀
DBIN ₁
DBIN ₂
DBIN ₃
DBIN ₄
DBIN ₅
DBIN ₆
DBIN ₇

⇒
 a fisare
 conect

1
 00000001
 00000010
 00000011
 00000100

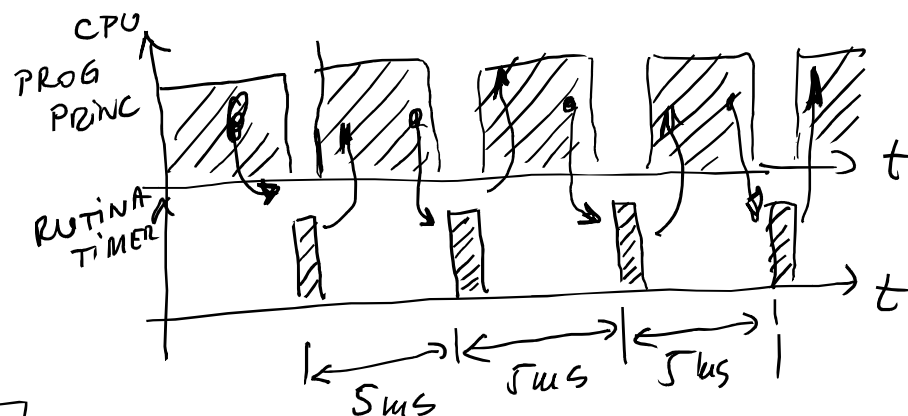
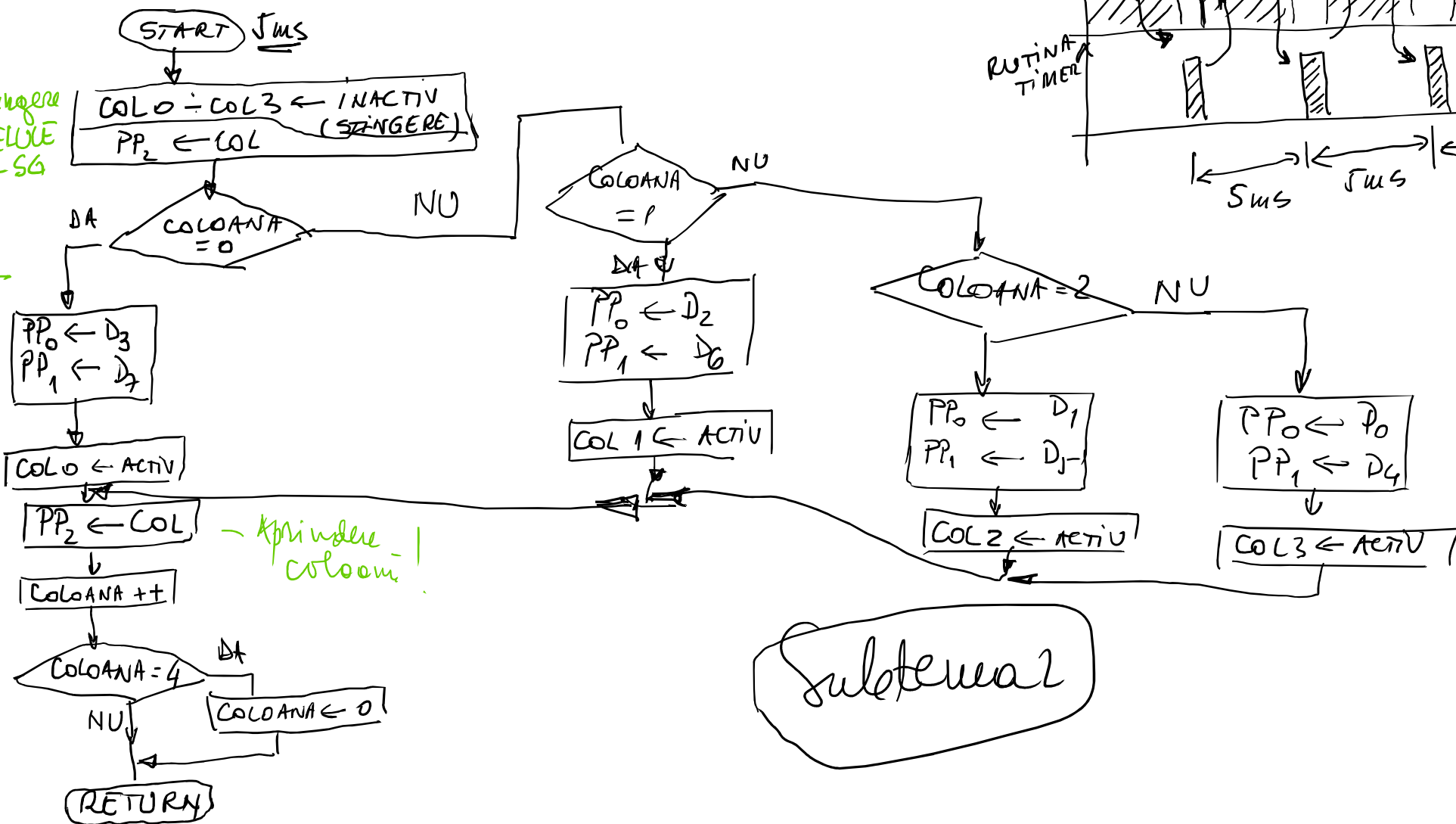
INC
 DIGIT + 1

COL ⇒ COL 0, COL 1, COL 2, COL 3

RUTINA TRATARE
INTERRUPTURE TIMER
(SEMNAUZARE TRECERE 5ms)

stingere
celule
7 SA

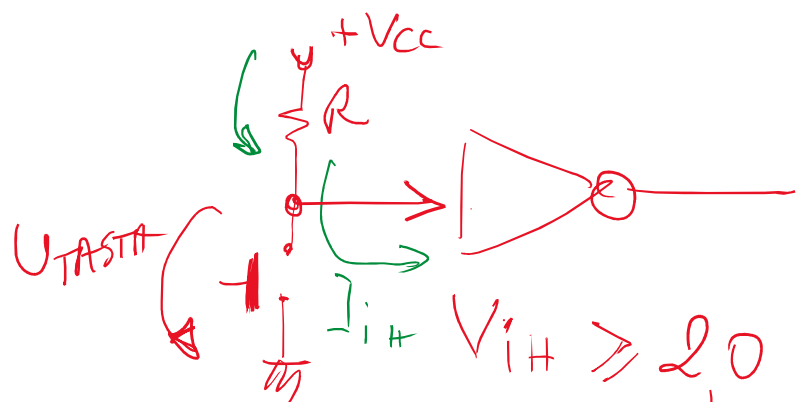
aprinde
coloani!



subteuara

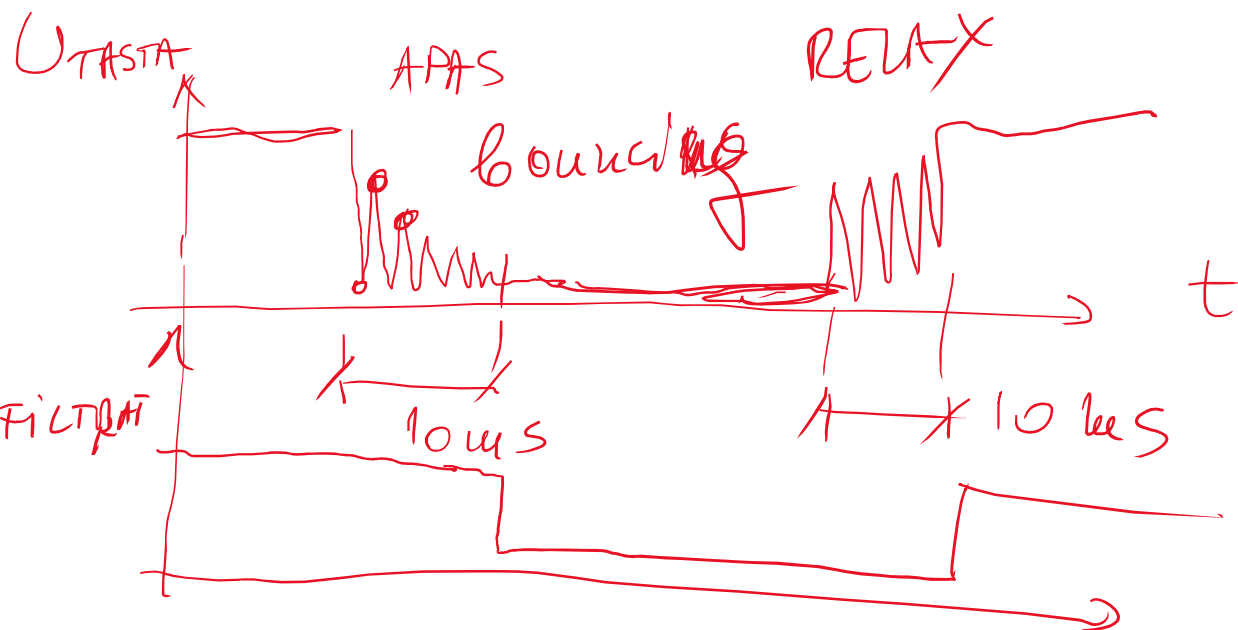
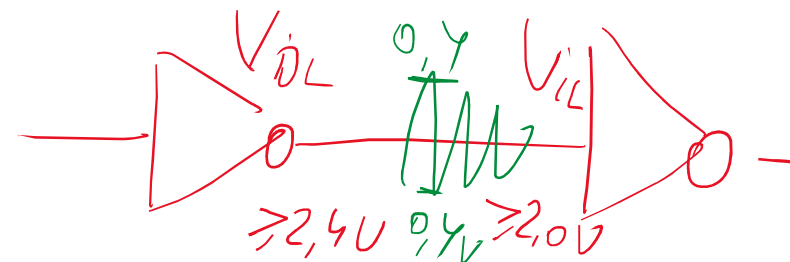
Nume variabila	Semnificatie	Valoarea de initializare	Mod de reprezentare	Adresa de stocare in memorie RAM
D0-D7	Imagini SW format 7 sg pentru celulele LED	7f(h) – CC; 00(h) – AC (segmente aprinse)	octet (8) – 7 biti utili / octet	30(h) – 37(h)
DBIN0-DBIN7	Imagini SW format binar (BCD) pentru celulele LED	08(h)	octet (8) – 4 biti utili / octet	38(h) – 3f(h)
COL0-COL3 (COL)	Imagini SW pentru comenzi pe coloanele comune	01(h) – AC; 0e(h) - CC	4 biti – accesibili individual	20(h) – zona adresabila pe Bit (00h, 01h, 02, 03h)
COLOANA	Indica ce coloana se actioneaza	03 – 00 (4 valori)	Octet	40(h)

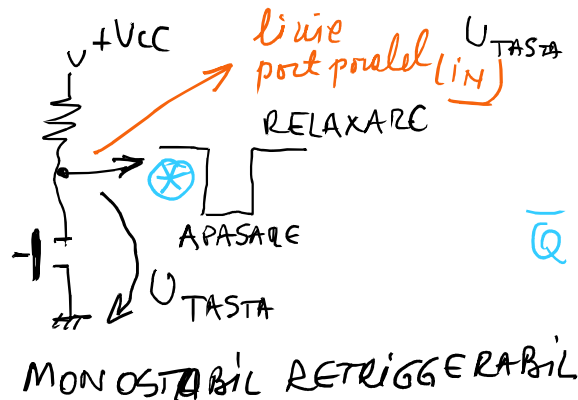
END 11 mar 2026



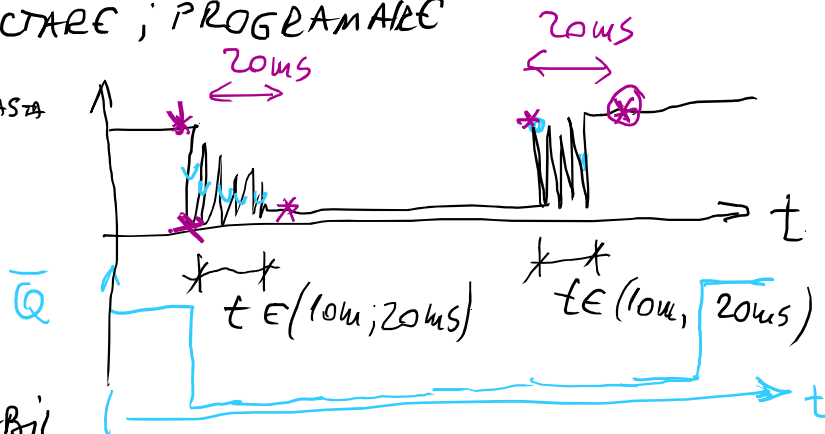
$$V_{IH} \geq 2,0V - V_{OH} \geq 2,4V$$

$$V_{IL} \leq 0,8V - V_{OL} \leq 0,4V$$

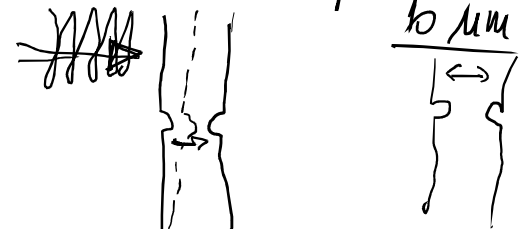
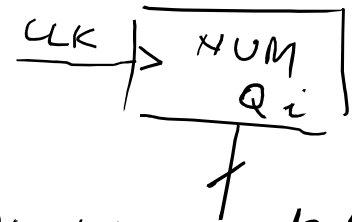




MONOSTABIL RETRIGGERABIL



bounce
debouncing



20kV - 1 м
5v - 250 μH

folosi're
optimale'
linii PP

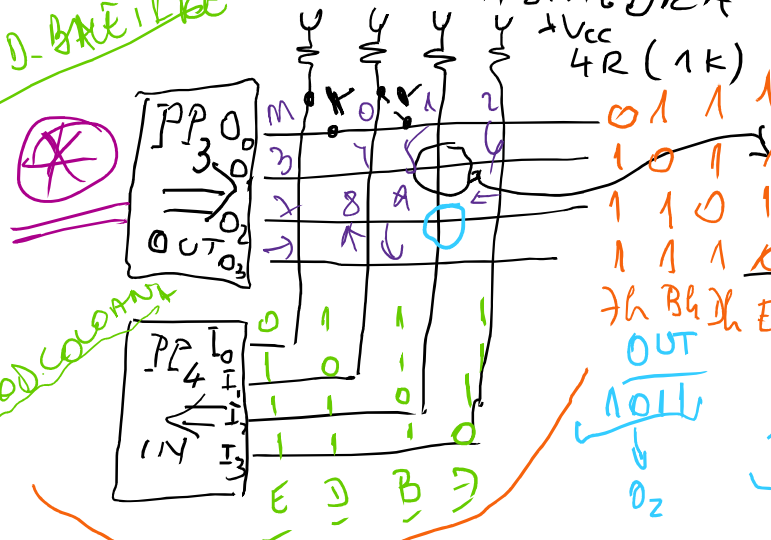
14 TASTE

$$2 \times 7 = 14$$

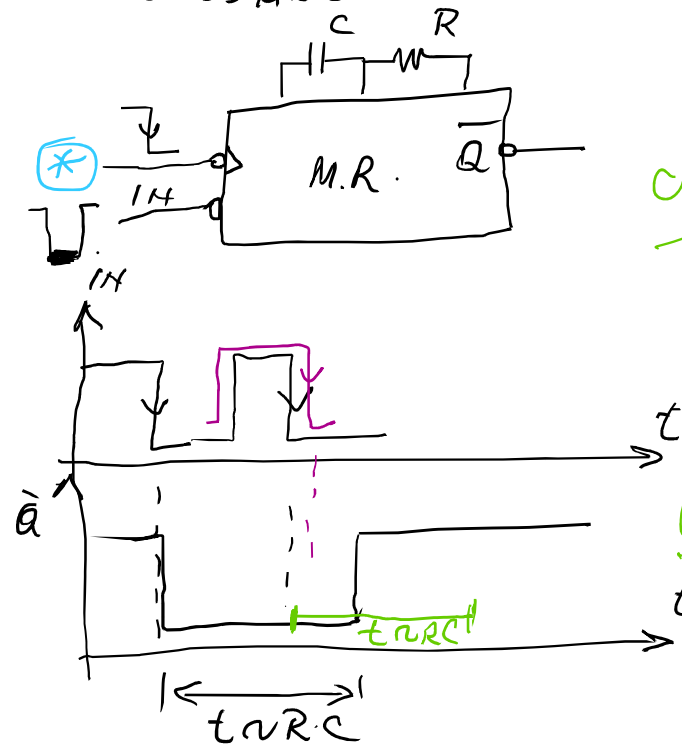
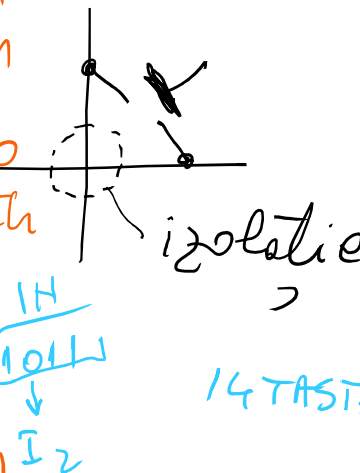
$$4 \times 4 = 16$$

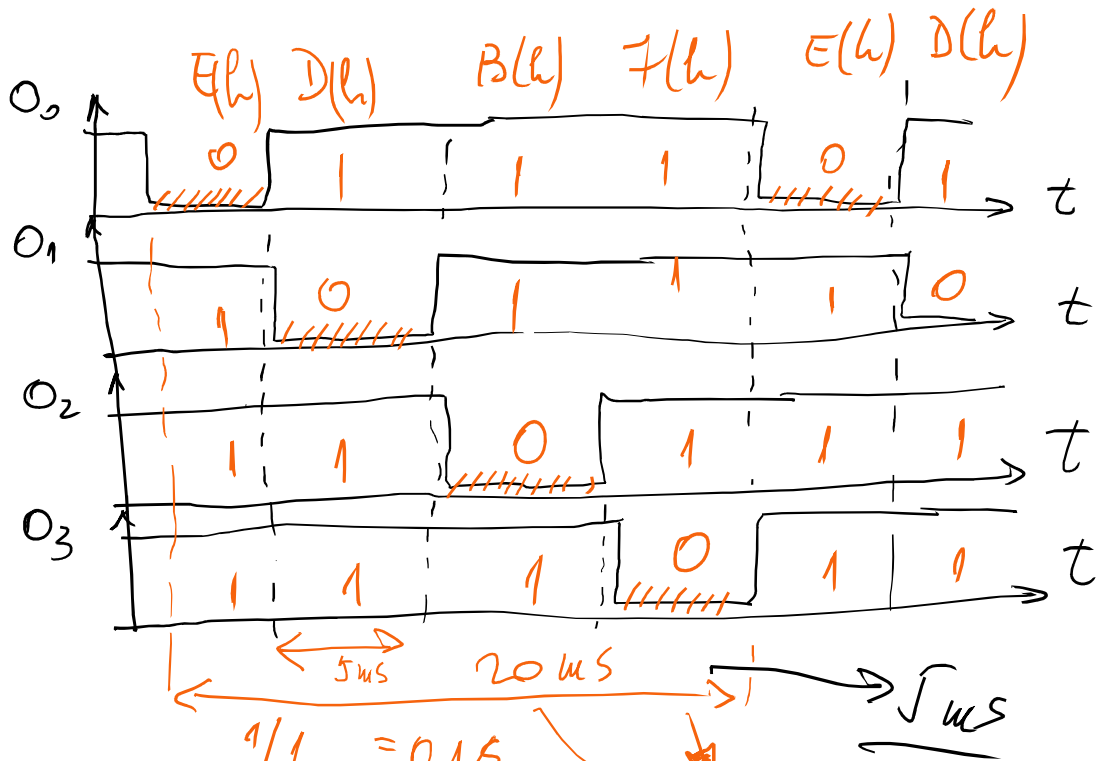
COMANDA MULTIPLEXATA
TASTATURA

COD-Beispiele



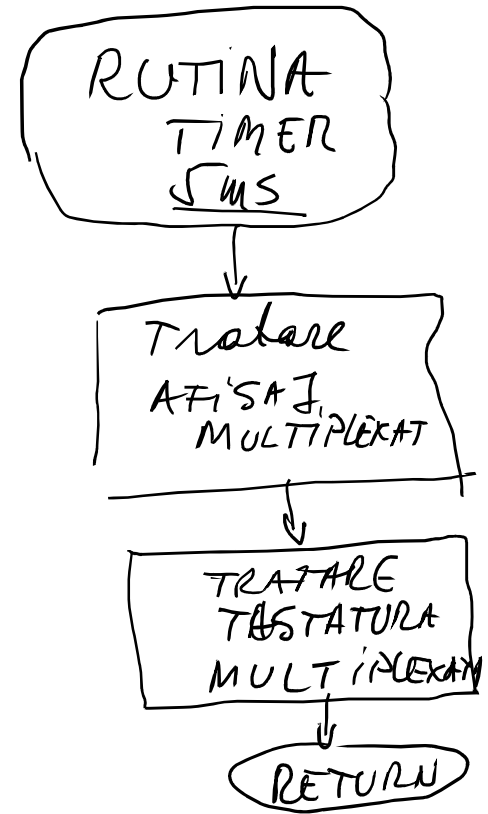
SUBTEMA 3 (PARTE HW TASTATURA)





Proces de baleiere a tastaturii -

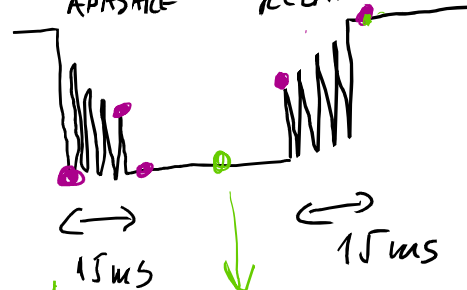
$E \rightarrow D \rightarrow B \rightarrow 7 \rightarrow E$



$$1/10 = 0,15$$

(100ms)

APASARE SI RELAXARE TASTA



APAS | TASTA_APAS | RELAX |

TIMER 5ms

TRATARE
AFISAJ
MULTIPLEXAT

TRATARE
TASTATURA
MULTIPLEXATA

RETI

SEMAFOR_TASTA

- indica
tasto
apreso
corect

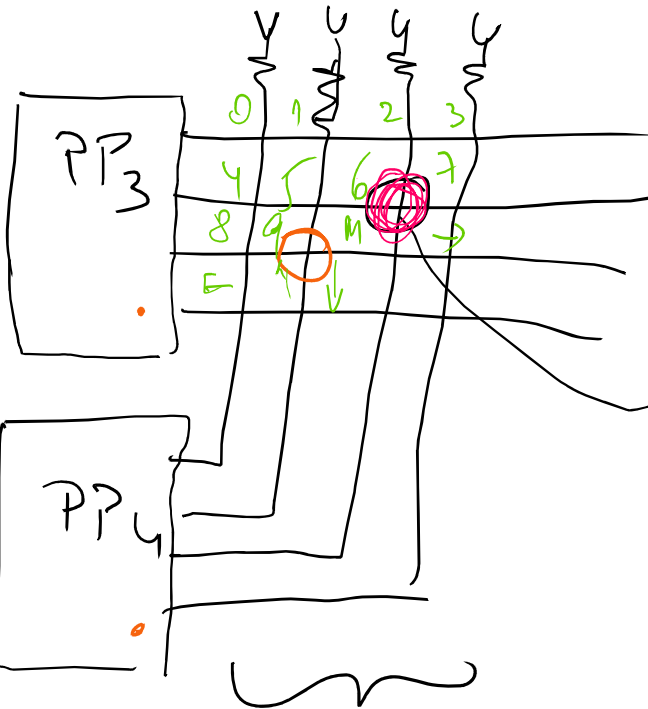
COD_BALEIA]

COD_TASTA_COL

} tasto
apreso

=> progr.
principal

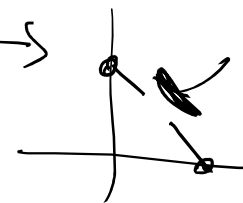
cod
baieia
 $\in \{07h, 08h, 09h, 0Eh\}$



COD_TASTA_COL
(4b)

locus

COD_BALEIA]
(4b)



identificator tasto
apreso



COD_BALEIA]
(4b)

COD_TASTA_COL
(4b)

1 0 1 1
B

1 1 0 1
D

0
B Dh

07h 08h
0 1 1
1 0 1
1 1 0
1 1 1

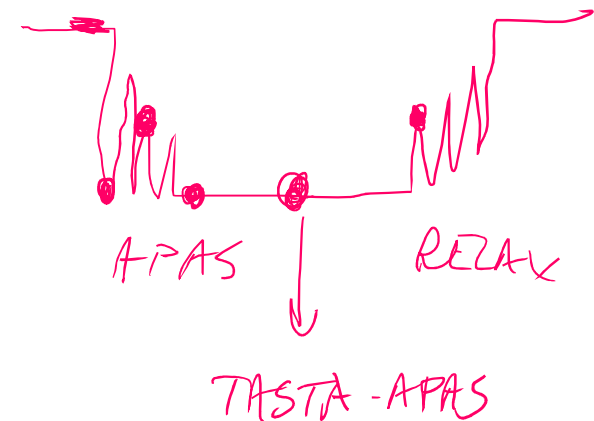
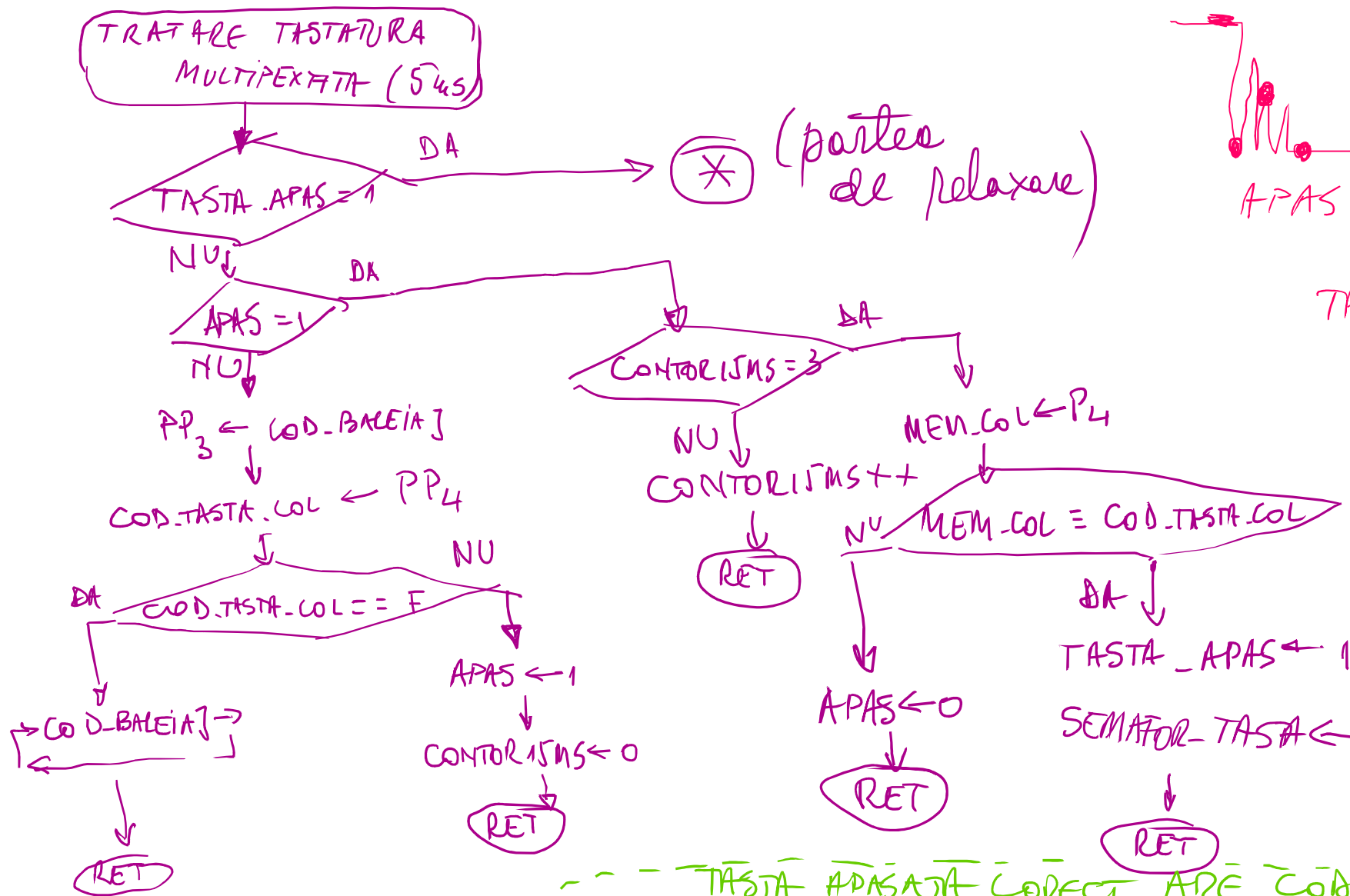
⊗ La lucru cu o tastatură este necesar:

- A - identificarea momentului apăsării unei taste
- B - realizarea unei temporizări (15 μs) pentru evitarea bouncing-ului
- C - verificarea stării de apăsare a tastei respective (0)
 - identificarea momentului relaxării unei taste
 - realizarea unei temporizări (15 μs) pentru evitarea bouncing-ului
 - verificarea stării de relaxare a tastei (1)

⊗ Prelucrarea unei taste apăsată de către o altă componentă este necesară:

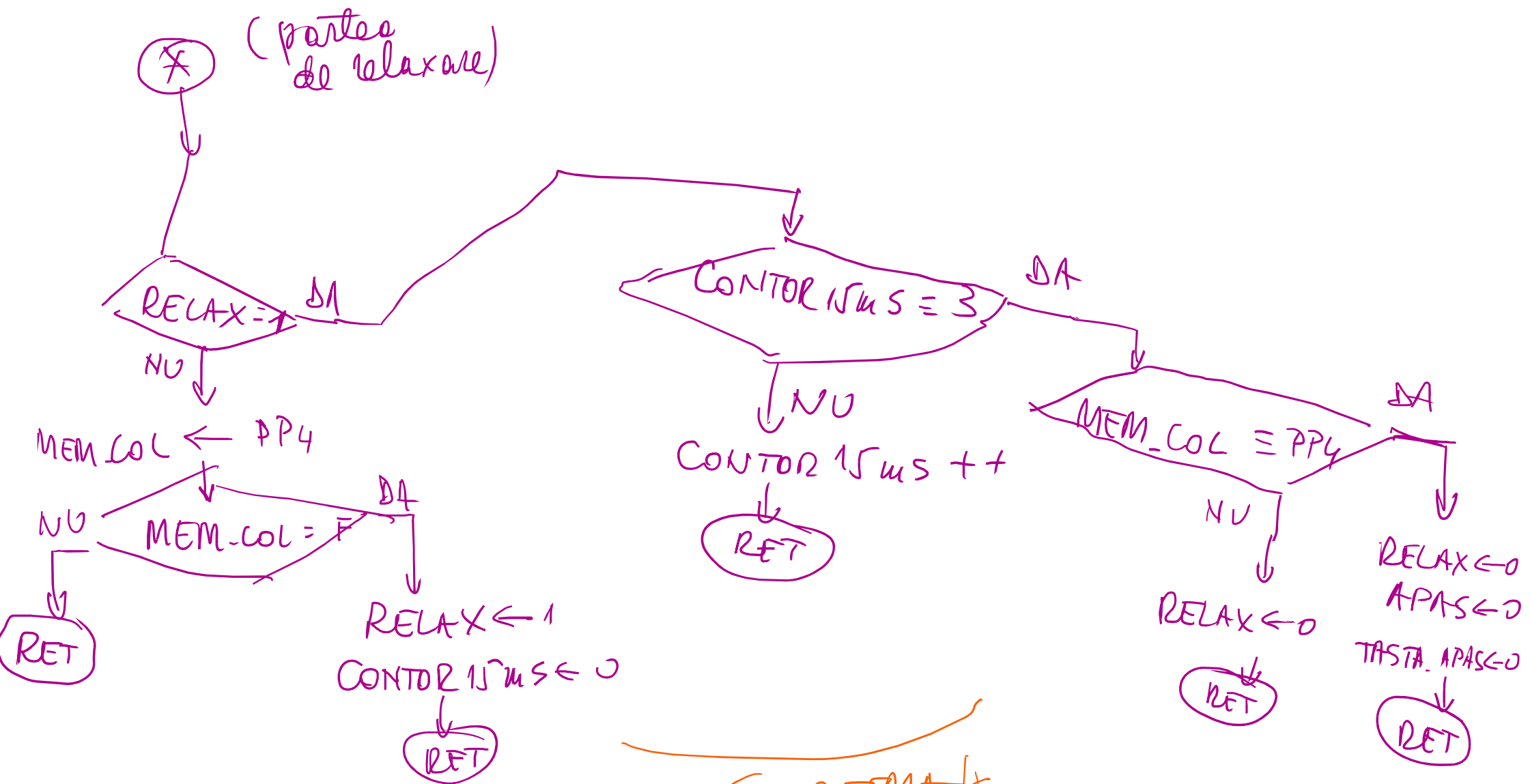
- semnalizarea unei taste corect apăsată (prin A, B, C)
1 variabilă SEMAFOR-TASTA
- Codul de identificare a tastei apăsată (variabilele obținute prin concatenare cod de baleiere-pe linii și cod citit coloană)
COD - BALEIAZ COD - TASTA - COL

END 18 mar 2026



--- TASTA APASATA CORECT ARE CODUL DE IDENT.
FORMAT AIN COD.BALEIA] SI COD.TASTA.COL !

INBICA
CATRE
PROG.PRINC
APASARE
CORECTA TASTA!



SUBTEMA 4

(SW - TASTATURA MUX.)

Manual de utilizare al tabelor

Subtema 5

- 1) Tabelul are 2 moduri de funcționare
- 1) mod 0 - operare normale

2) mod 1 - programare

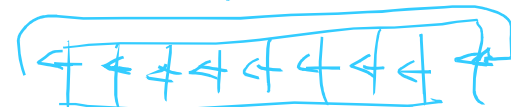
Modul se scrie prin apăsarea tastei M

- 2) În modul operare normale

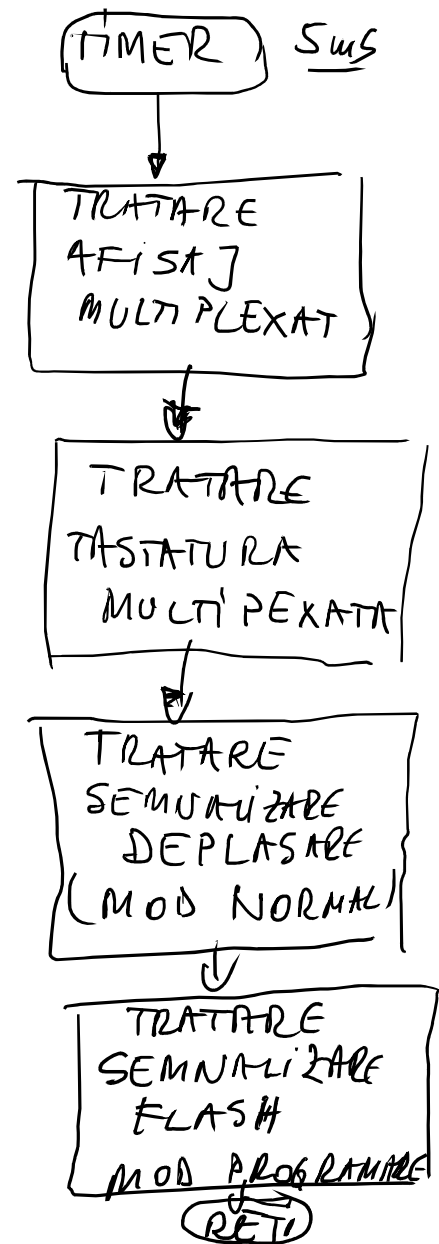
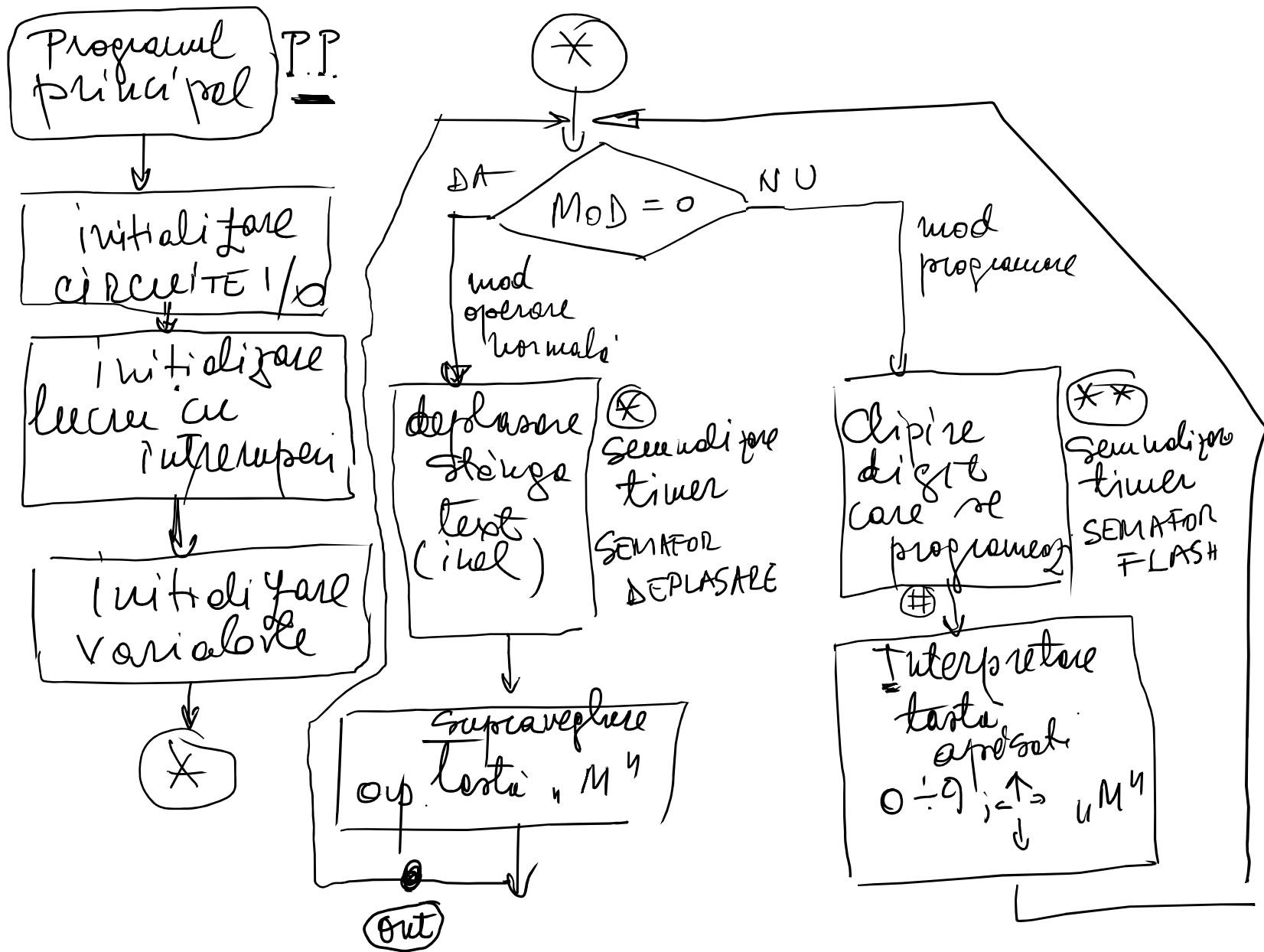
- textul afișat se deplasează
- cifre stâng și dreapta viteze de deplasare indicate în funcție
- supraveghere taste M
- și curenții mod
- nici o altă tastă nu acționează în modul 0 !

- 3) Modul programare
- = textul afișat rămâne fix
- = primul digit (dreapta) va clipi
- Cu timpul indicat în funcție
- = supraveghere taste apăsate
- 0, ..., 9, $\leftarrow$ $\uparrow$ $\rightarrow$, M
- = La tastele 0 ÷ 9 $\rightarrow$ se programează digitul cu valoarea

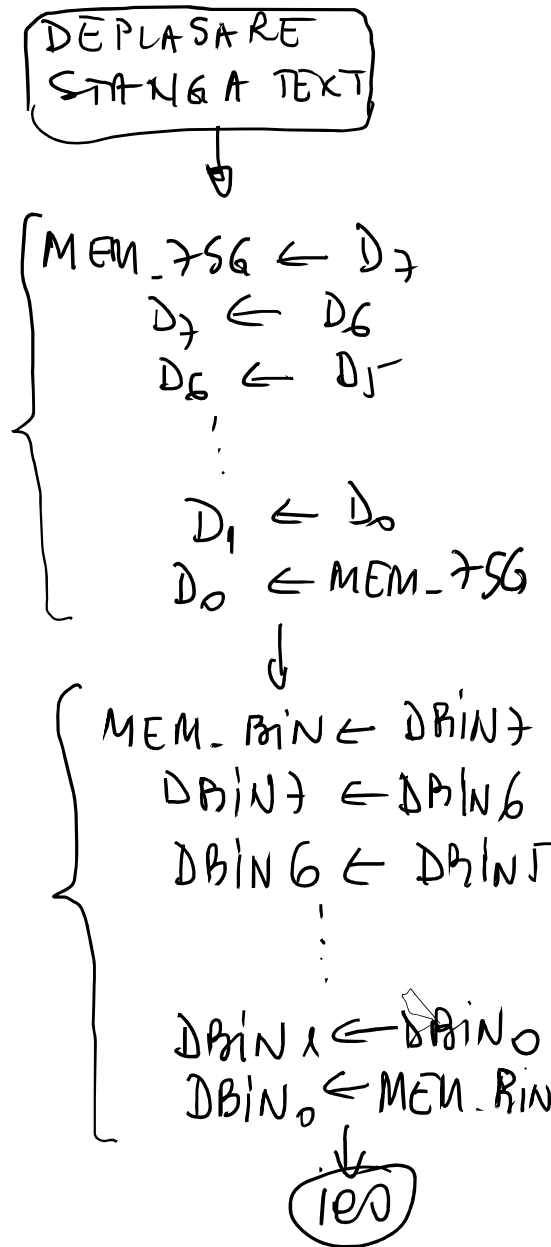
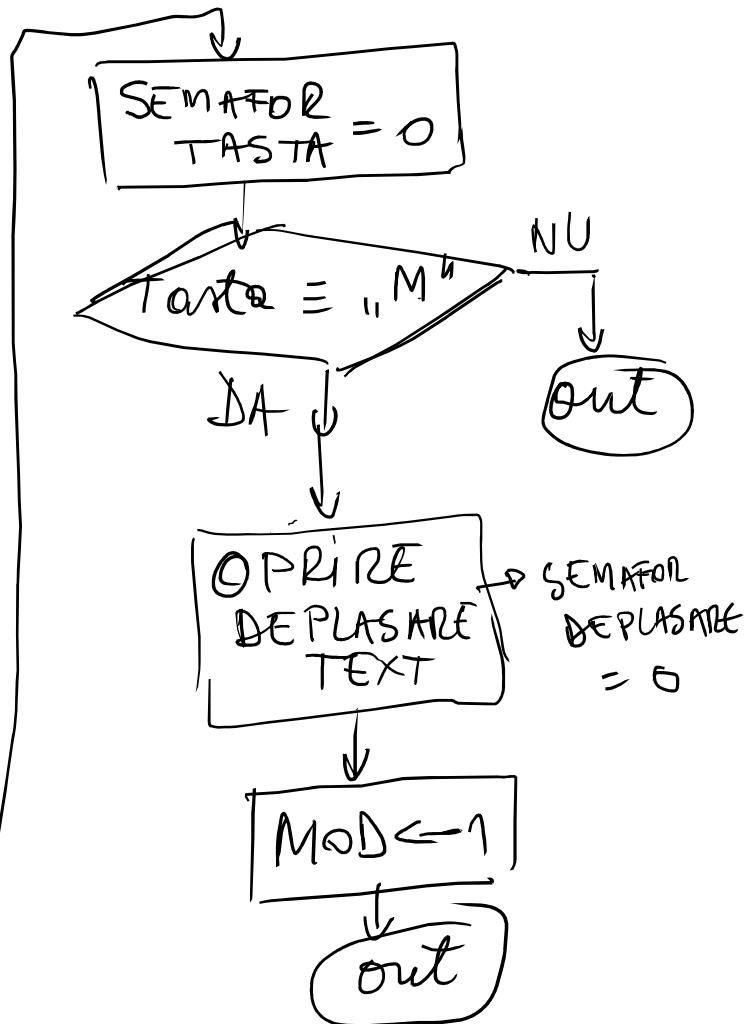
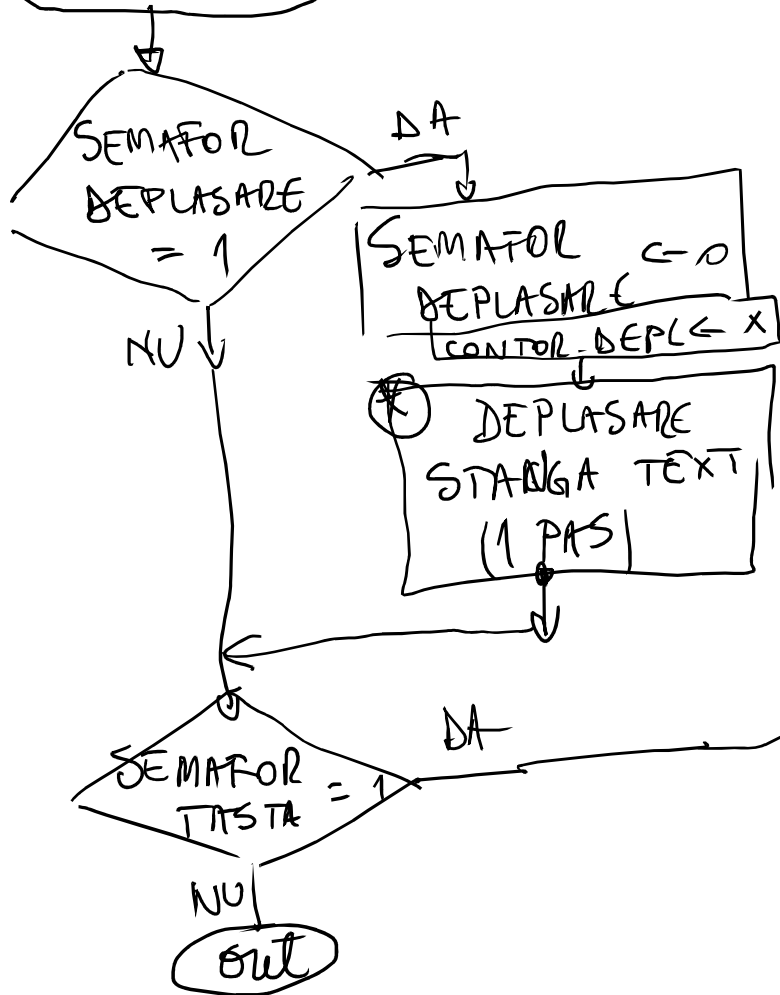
- = cu tastele $\uparrow$ - incrementare
- $\downarrow$ - decrementare
- = cu tastele $\leftarrow$ $\Rightarrow$ aleg următorul digit care se programează $\leftarrow$

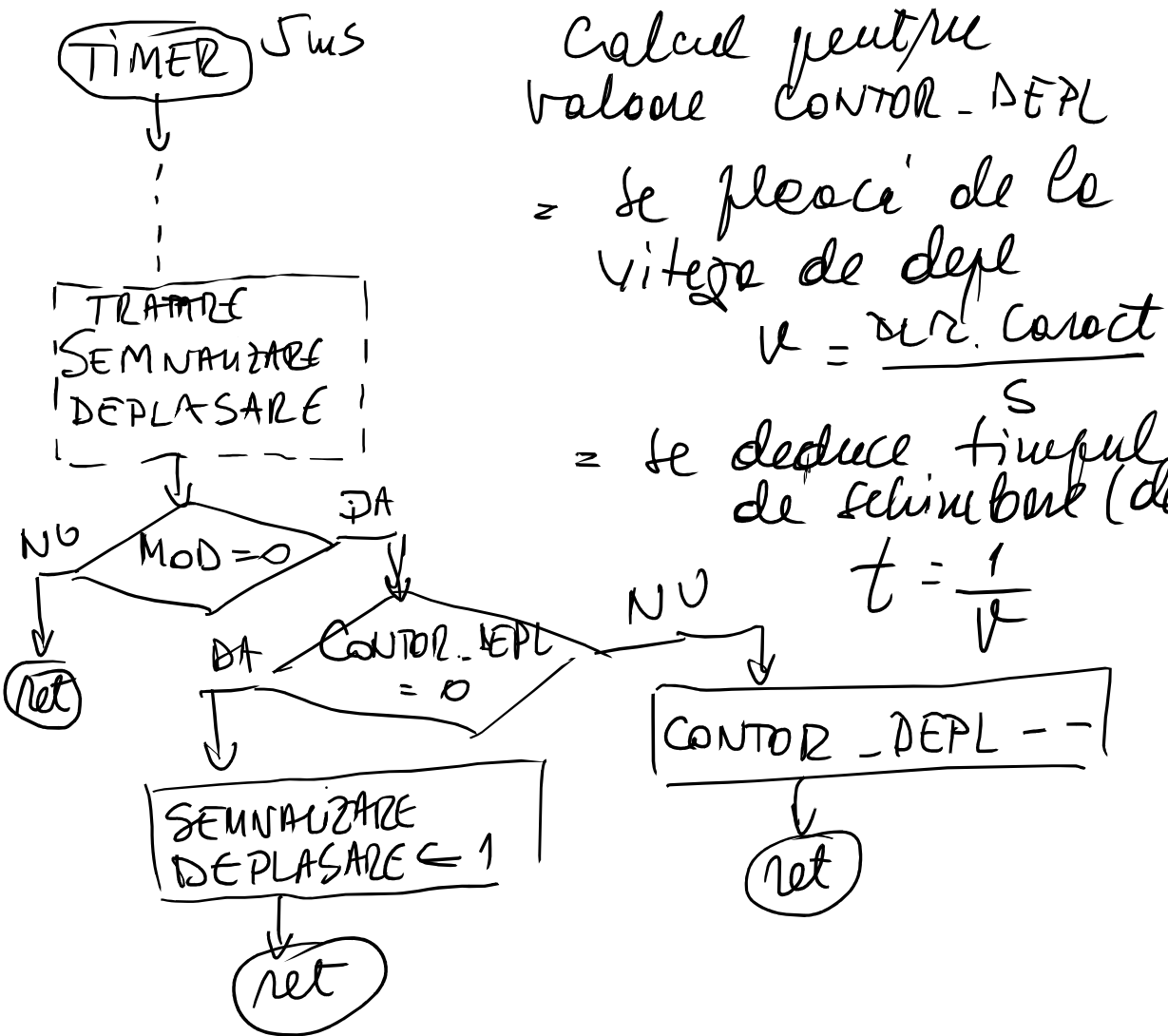


END 25 mar 2026



MOD
OPERARE
NORMALA
(din P.P.)





Calcul pentru
valoare CONTOR-DEPL
= se pleacă de la
viteza de depl
 $v = \frac{nr. \text{caract}}{s}$

= se deduce timpul
de schimbare (deplasare)
 $t = \frac{1}{v}$

= valoarea
CONTOR-DEPL
 $= \frac{1}{v}$
 $\frac{5ms}{0.005s}$
etalonul
caz care maror
timpul

PROBLEMA

- verificarea operații
cu variabila CONTOR-DEPL

ex 4 caract/s
 $v = 4 \frac{1}{s}$
 $t = \frac{1}{4}$
 $CONTOR = \frac{1.000}{4 \cdot 5} = 50 \text{ (V)}$

END 1 apr 2026